

# Photo Organizer Tools

## A Simple Guide to Cleaning Up Your Photos

Source code: [github.com/LoganCalloway/photo-organizer](https://github.com/LoganCalloway/photo-organizer)

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These three tools help you clean up a messy photo collection. They find duplicate photos, sort them by date, and remove overlapping photos between two folders. Everything is done safely — nothing is permanently deleted until you decide to delete it yourself.

**Important:** Always make a backup copy of your photos before running any of these tools. Better safe than sorry!

## Requirements

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Before you start you need two things installed on your Mac:

1. **Python 3** — comes pre-installed on Mac. To check, open Terminal and run:

```
python3 --version
```

2. **ExifTool** — needed only for the sorting tool. Install it with:

```
brew install exiftool
```

If **brew** is not found, install Homebrew first at [brew.sh](https://brew.sh), then run the command above.

## How to Open Terminal

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Press **Command + Space**, type **Terminal**, and press Enter. This is where you will type all the commands in this guide.

## Tool 1 — Find and Remove Duplicates

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**File:** `find_duplicates.py`

This tool scans your photos, finds exact duplicate files, and moves the extra copies to a **Duplicates/** folder. It keeps the best copy of each photo and leaves everything else untouched.

**Step 1 — Preview (nothing moves)**

```
python3 find_duplicates.py /path/to/your/photos
```

This shows you exactly what it would do without moving anything. Read through the output and make sure it looks right.

## Step 2 — Run it for real

```
python3 find_duplicates.py /path/to/your/photos --execute
```

Duplicate photos are moved to a **Duplicates/** folder inside your photos folder. **Nothing is permanently deleted** — you delete the Duplicates folder yourself when you are happy.

### How it works

- Reads every photo and creates a unique fingerprint (MD5 hash) based on the file contents
- Groups photos with the same fingerprint together — these are exact duplicates
- Keeps the copy that is in the most specific subfolder
- Moves all other copies to Duplicates/

### Example output

```
KEEP:    Pictures/Christmas 2018/IMG_4497.jpg
REMOVE:  Pictures/Backup/IMG_4497.jpg (2.1 MB)
REMOVE:  Music/Pictures/IMG_4497.jpg (2.1 MB)

Duplicate groups:  1,203
Files to remove:   2,847
Space to free up:  36.4 GB
```

## Tool 2 — Sort Photos by Year and Month

**File:** sort\_photos.py

This tool reads the date each photo was taken and moves it into a folder organized by year and month.

### Step 1 — Preview (nothing moves)

```
python3 sort_photos.py /path/to/photos /path/to/output
```

### Step 2 — Run it for real

```
python3 sort_photos.py /path/to/photos /path/to/output --execute
```

Replace `/path/to/photos` with your photos folder and `/path/to/output` with where you want the sorted folder created.

## Result

Your photos will be organized like this:

```
Photos-Sorted/  
  2018/  
    06 - June/  
    12 - December/  
  2019/  
    03 - March/  
  Unknown/      <- photos with no readable date
```

## How it works

- Reads the EXIF date embedded inside each photo — the actual date it was taken
- Falls back to the file modification date if no EXIF data exists
- Photos with no detectable date go into an **Unknown/** folder so nothing is lost
- Moves photos into the new sorted folder

## Tool 3 — Remove Overlapping Photos Between Two Folders

**File:** `remove_overlaps.py`

Use this when you have two separate folders and some photos exist in both. It removes the copies from one folder while leaving the other completely untouched.

### Step 1 — Preview (nothing moves)

```
python3 remove_overlaps.py /path/to/folder1 /path/to/folder2
```

### Step 2 — Run it for real

```
python3 remove_overlaps.py /path/to/folder1 /path/to/folder2 --  
execute
```

**folder1** is the folder you want to clean up — duplicates will be removed from here. **folder2** is the folder you want to keep intact — it is never touched. Photos found in both folders are moved out of folder1 into a **Duplicates-[folder2 name]/** folder for review. The folder names in the output will automatically match whatever folders you provide.

## How it works

- Scans both folders and fingerprints every photo
- Finds photos that exist in both folders

- Moves the copies out of folder1 — folder2 is never touched
- Removed copies go to a review folder, not permanent deletion

## Tips and Notes

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- **Always preview first** — run without `--execute` to see what will happen before committing
- **Make a backup** — copy your photos to an external drive before running anything
- **Sign out of cloud sync** — if your photos are in OneDrive or iCloud, sign out before running the scripts so nothing gets synced mid-process
- **Near duplicates** — these tools only catch exact byte-for-byte duplicates. Photos that look the same but were re-saved or sent via text may not be caught
- **Scanned photos** — old scanned photos have no EXIF date. Use the `touch` command to set their date manually before sorting:

```
find /path/to/scanned/photos -exec touch -t 198001010000 {} +
```

Then run ExifTool to embed the date into the files:

```
exiftool -DateTimeOriginal="1980:01:01 00:00:00" -CreateDate  
="1980:01:01 00:00:00" -r /path/to/scanned/photos  
find /path/to/scanned/photos -name "*_original" -delete
```

## Links

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- **Source code:** [github.com/LoganCalloway/photo-organizer](https://github.com/LoganCalloway/photo-organizer)
- **Homebrew:** [brew.sh](https://brew.sh)
- **ExifTool:** [exiftool.org](https://exiftool.org)